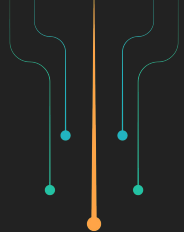




QLF DECENTRALIZED TRUST

Ethereum Ecosystem Briefing

[Book a virtual briefing >](#)



Linux Foundation Decentralized Trust

The open source foundation for decentralized
technology ecosystems

LFDT is the neutral, Linux Foundation-governed home for enterprise-grade Ethereum and decentralized trust infrastructure, where projects like Besu anchor real-world adoption across tokenized money, assets, identity, privacy, and interoperability, enabling regulated institutions to move from pilots to production with open governance, client diversity, and long-term resilience.

[About us >](#)

A part of the Linux Foundation

By embracing the open source principles established by the Linux Foundation, LF Decentralized Trust serves as the neutral home for the open development of technologies that empower organizations to innovate with secure and resilient code.

[Visit Linux Foundation >](#)



The future of decentralized technologies

open source

open development

open governance

“

Open source and open development are at the core of what we do. By working collaboratively under open governance principles, we ensure that the technology is transparent, secure, and continuously improved by a diverse group of contributors.

Daniela Barbosa - Executive Director, LF Decentralized Trust

10

Years
Community
Building

100s
Global

Enterprises &
Startups

7

Graduated
Projects

11

Incubating
Projects

50+

Labs

1

Standards
Project (JDF)

7

Regional
Chapters

186

Meetup
Groups

110+

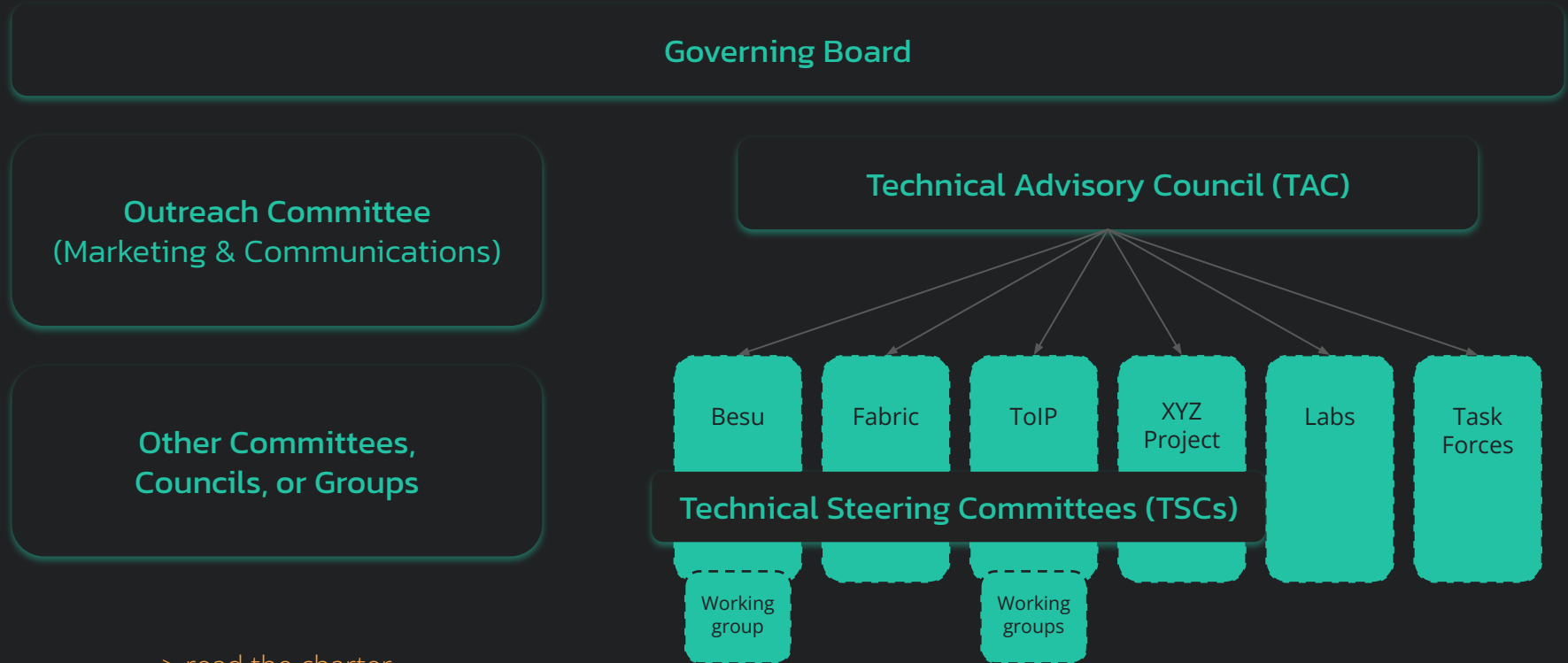
Meetup
Participants

200k

Subscribers

Growth & Momentum

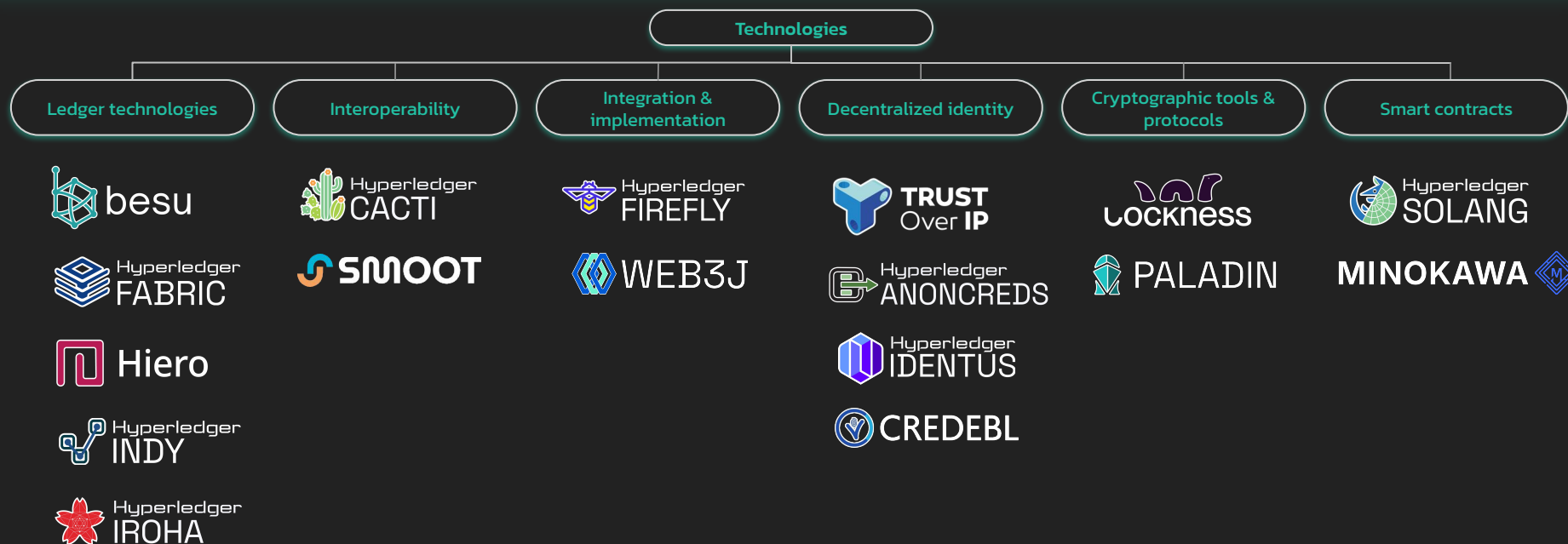
LF Decentralized Trust governance structure



> [read the charter](#)

[Project TSCs reps can be viewed in project Github](#)

A growing landscape of decentralized technologies



[View projects >](#)

Spectrum of blockchains - for today's enterprise

Not one size fits all!

Permissioned vs. Permissionless: Who can write to a Blockchain (accessibility)

Public vs. Private: Who can read from a Blockchain (visibility)

Permissionless Public

Permissionless Private

Permissioned Public

Permissioned Private



Bitcoin, Ethereum,
Solana, Cardano,
Hedera, +++



Private
Payments

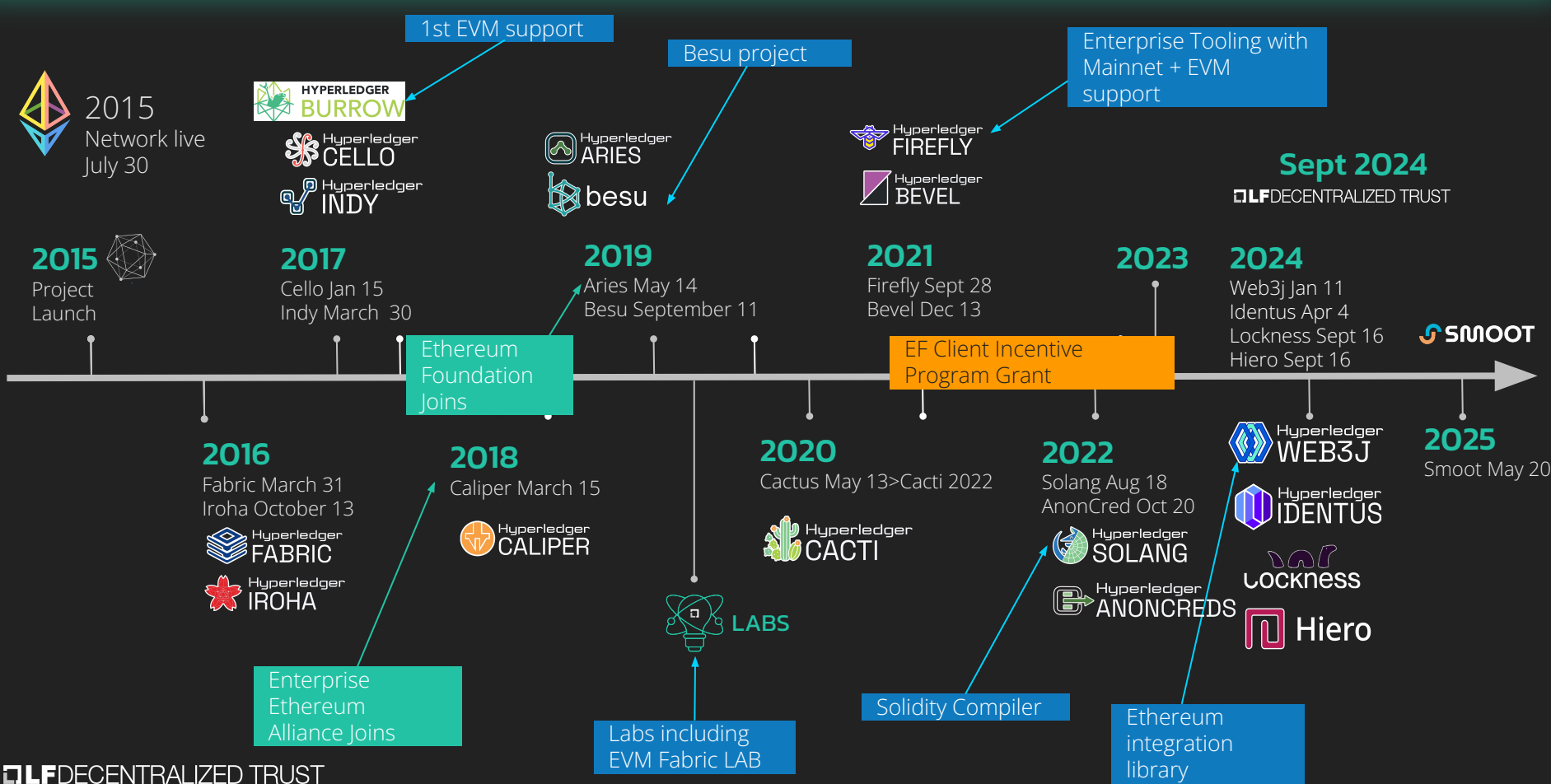


Land titles, University degrees



Medical records

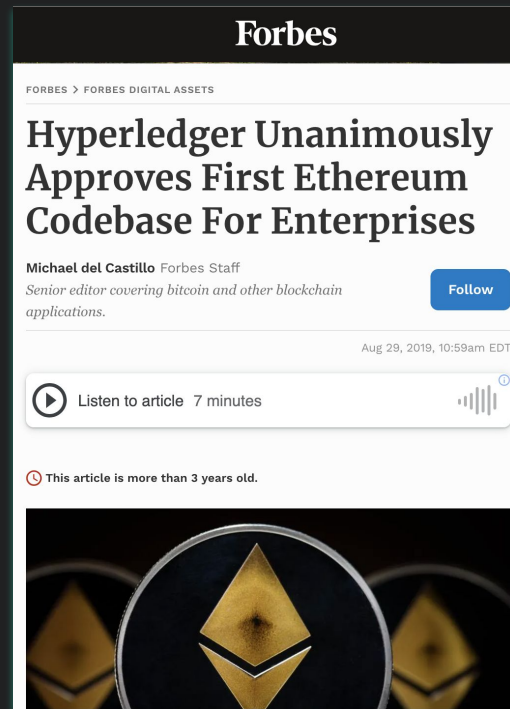
LF Decentralized Trust and Ethereum ecosystem







- ❑ Contributed by Consensys Aug 2019
- ❑ Open source Ethereum execution client written in Java
- ❑ Public and private options
- ❑ Implements the EEA specification
- ❑ Can run all Ethereum smart contracts and works with all Ethereum ERC Standards, including:
 - ❑ ERC-20 - Fungible Tokens
 - ❑ ERC-721 - NFT Tokens
- ❑ Permission EVM with full parity with mainnet
- ❑ Besu is being used by many in the financial industry as well as those that **wish to work in both the public and private blockchain space** or those that don't want to have to make a firm decision between working on public or private chains.



Besu : 2025 Accomplishments and 2026 Roadmap

2025 Accomplishments

- Enabled Ethereum gas limit increases from 30M to 60M through worst case opcode and precompile optimizations.
- Faster sync with reduced indexing, improved RLP decoding, and optimized Snap DB heal.
- Better RPC throughput and debug tooling.
- General execution gains with parallel processing, global code cache, and improved non-finality handling.

2026 Focus

- Raising mainnet gas target toward 100M.
- Bonsai archive performance for L2 adoption.
- More sync and EVM arithmetic improvements with a native UInt256 type.
- Productionizing Tuweni upgrades for faster core byte operations.



[Read blog](#)

Client diversity is **NOT** optional: Besu is critical

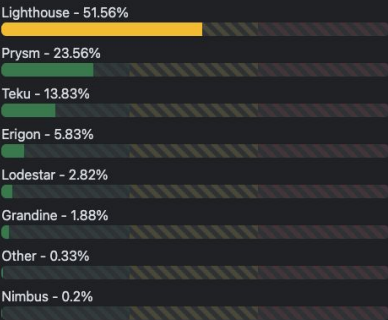
Client Distribution

View Staking Pool Diversity

Goal: <33% | Danger: >66%

Consensus Clients

⚠ Lighthouse has a majority, switch to a minority client!



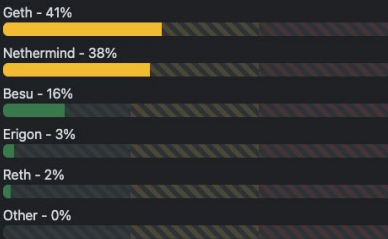
Data provided by [Miga Labs](#) — updated daily.
Data may not be 100% accurate. ([Read more](#))

Data source ([read more](#)):

● Miga Labs ● Rated.Network

Execution Clients

⚠ Switch from Geth, Nethermind to a minority client!



Data provided by [supermajority.info](#) — updated manually.
Data may not be 100% accurate. ([Read more](#))

Based on 73.9% self-reported network coverage with the remaining assumed to be mostly Geth.

Data source ([read more](#)):

● Supermajority.info ● Ethernodes

Besu has ~16% share of all Ethereum mainnet nodes.

<https://clientdiversity.org>

A recent analysis by Blockdaemon of the technology powering the global tokenization ecosystem found that Besu was the most frequently mentioned protocol in a survey of 92 different firms, ranging from bank asset managers to clearing houses.

40% of firms surveyed cited Besu

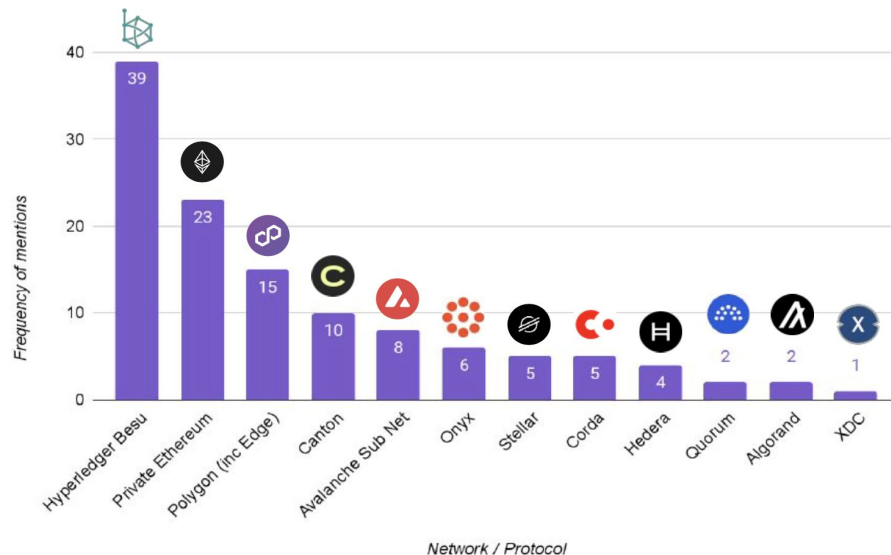


Figure 3 - Tokenization protocol leaderboard

The Next Phase of Enterprise Ethereum Infrastructure

Lean Core

Besu is evolving toward a lighter core runtime focused on the EVM specification, improving maintainability, performance, and long-term sustainability

Pluggable Architecture

Enterprise and advanced capabilities are moving into modular plugins, allowing features to evolve independently and be deployed based on network requirements

Enterprise Runtime

Enterprise deployments combine the Besu core + selected plugins into a single runtime tailored for institutional networks.

DLF DECENTRALIZED TRUST

Besu's Journey: Bringing
Ethereum into the Enterprise



[Read blog](#)

Multi purpose enterprise Ethereum

Besu serves as an **execution client** on **public proof-of-stake Ethereum networks**, including **Ethereum Mainnet, Goerli, and Sepolia**. It can be run as an execution client with any consensus client on Ethereum Mainnet. Currently, **~16% of Mainnet execution clients run Besu.***

Besu is used in Layer 2 (L2) solutions, which are secondary frameworks or **protocols designed to enhance scalability and transaction speed** on Ethereum. **L2s such as LINEA by Consensys** leverage Besu to optimize performance and efficiency.

Blockchains connected to Ethereum Mainnet via a two-way peg enable **secure asset transfers** while operating with **different consensus mechanisms**. The **Asian Development Bank Cross-Border Securities Transaction System** leverages this approach for enhanced interoperability.

Combines public accessibility with permissioned governance, Besu is well-suited for applications requiring **transparency, security, and regulatory compliance**. Public infrastructures such as LACCHAIN, EBSI (Europeum), Rede Blockchain Brasil, and Alastria leverage Besu to meet these requirements.

Besu offers comprehensive permissioning schemes tailored for consortium environments, ensuring controlled access to participants and transactions. It is **widely used in private blockchains for settlement systems, trade finance, and cross-border payments**. Central banks in Brazil, Norway, and Thailand have leveraged Besu in their CBDC adoption initiatives.



Mainnet



L2 Networks



Hybrid Interop



Public
Permissioned



Private
Permissioned



besu



Financial services adoption and leadership

Besu Financial Services Working Group

The mission is driving working sessions for enterprise users and code contributors to come together for:

- Identifying challenges that enterprise users face when operationalizing Besu for enterprise use cases
- Defining new features and enhancements that drive alignment between Besu and enterprise-level requirements

The main deliverables expected to come out of this WG include, but are not limited to:

- A prioritized and detailed list of enhancements of existing features of Besu
- A prioritized and detailed list of new features of Besu
- A prioritized and detailed list of technical limitations of Besu
- A list of recommended practices of Besu for enterprise use cases

accenture

DTCC

citi

consensys

kaleido

BANCO CENTRAL
DO BRASIL

JPX
JAPAN SECURITIES
CLEARING CORPORATION

mastercard

LACCHAIN

Santander

Vanguard

VISA

Web3 Labs

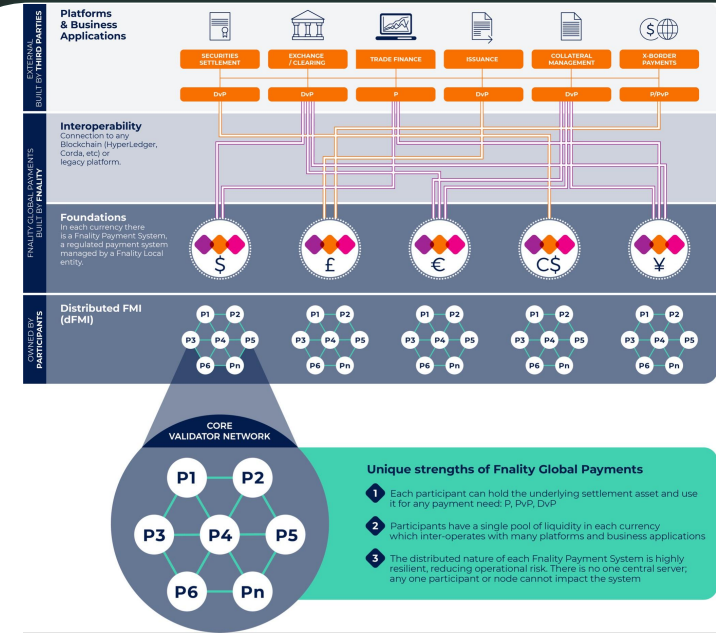
[Visit working group >](#)



Private Permissioned Financial Services Use Cases



- Fnality is distributed Financial Market Infrastructure built on Besu that interoperates and connects to many existing systems and rails
- Collateral assets are tokenized and pooled among nodes in the network for use across currencies, payment types, and even with other blockchain networks
- The network reduces systemic and credit risk by utilizing the dFMI infrastructure and enabling faster settlement among parties
- Fnality provides a digital representation of money held in a central bank account to ensure banks could rely upon token value as they would fiat currency in transactions



Fnality now boasts 17 major institutions as shareholders: Banco Santander, BNY Mellon, Barclays, CIBC, Commerzbank, Credit Suisse, Euroclear, ING, KBC Group, Lloyds Banking Group, Mizuho Financial Group, MUFG Bank, Nasdaq, Nivaura, Sumitomo Mitsui Banking Corporation, State Street Corporation, and UBS

Citi transforms transaction banking services with Besu

MEMBER COMPANY



TECHNOLOGY USED



SUMMARY

Early on, Citi saw blockchain's potential to address long-standing challenges in financial markets. So it isn't surprising that the organization began vetting distributed ledger technology (DLT) over a decade ago.

The announcement of Citi Token Services for Cash moving to a live commercial solution in October 2024 marked a turning point in finance.

RESULTS

- The platform has been connected to multiple Citi systems in the first year of its rollout
- Awarded the 2024 Model bank Award for Digital Asset Innovation by Celent, a global research and advisory firm for the financial services industry

[Read case study >](#)

Launched Jan 2023, during the World Economic Forum week in Davos, the Universal Digital Payments Network (UDPN) is a DLT-underpinned messaging backbone focused on providing interoperability between the fast-growing number of different regulated stablecoins and CBDCs, and seamless connectivity between any business IT system and regulated digital currencies.

The UDPN is a messaging network connecting all government-regulated digital currency systems, including regulated stablecoins and future CBDCs. The UDPN sandbox enables organisations to trial use cases in a controlled environment.

UDPN can help organisations access multiple digital currencies through a single locally-installed gateway.



UDPN is a global messaging network supporting government-regulated digital currency systems, including regulated stablecoins and Central Bank Digital Currencies (CBDCs). The UDPN promotes financial inclusion by allowing enterprises around the world to connect directly with the centralised and decentralised digital currency systems of the future.

Prototype digital platforms for green bond tokenization

Developed a digital infrastructure that can enable green investments and help issuers and governments to meet environmental and sustainability goals

Investors can both see accrued interest, and track in real time how much clean energy is being generated, and the consequent reduction in CO2 emissions. The investor can sell the bonds in a transparent market

Measurable Results include:

- Reducing the bond preparation and origination phase by an estimated 50% from 1-2 months to 2-4 weeks
- Streamlining the current subscription process and eliminating all currently manual reconciliation work can save an estimated 800-1,000 person in the subscription phase
- Saving an additional 2-3 business days when the green bond is listed for secondary market trading



HONG KONG MONETARY AUTHORITY
香港金融管理局



The prototype provides ledger optionality through an interoperability protocol that enables the communication between applications running on a Fabric or Besu network

> read report: [Project Genesis 2.0](#)



Mastercard MTN is a programmable payment platform built on Besu designed to unlock the full potential of digital assets in the modern payments ecosystem.

- Tokenizes customer deposits with features to mint, burn, and transfer tokens, to integrate bank money with digital assets.
- Enables banks to integrate with innovative applications through a single, secure gateway that simplifies both domestic and cross-border transactions.
- Uses advanced identity management and permissions (via Mastercard Crypto Credential) for trusted transactions.
- Offers a scalable, interoperable blockchain platform.



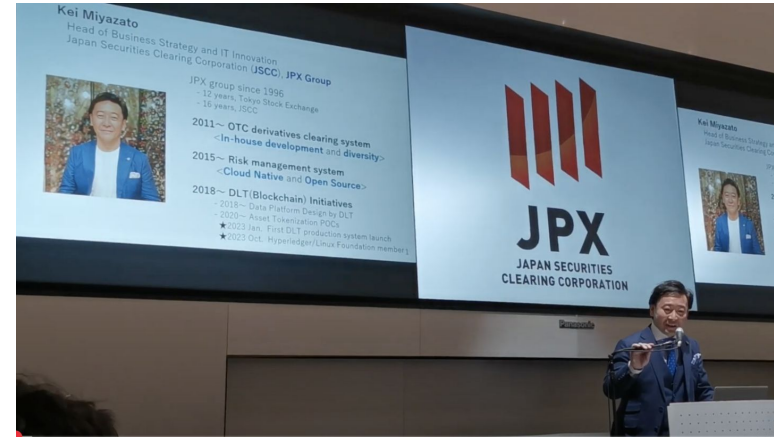
Mastercard MTN represents a strategic evolution in digital asset payments, leveraging Besu's capabilities to deliver secure, interoperable, and always-on financial services for banks, fintechs, and central banks globally.

Traditional settlement for commodity futures involves manual processes and physical delivery, leading to inefficiencies and risks, particularly during disruptive events like pandemic.

JSCC implemented a DLT-based settlement system using Hyperledger Besu to tokenize warehouse receipts and facilitate seamless cash settlements.

Key Benefits:

- Operational Efficiency – Reduced manual handling in the delivery process.
- Risk Mitigation – Minimized physical delivery risks.
- Future Expandability – Built on globally standardized DLT, Besu for scalability.
- Cost & Time Savings – Faster and more affordable settlements.



The Japan Securities Clearing Corporation (JSCC), a CCP within the JPX Group, has been actively exploring and implementing blockchain technology, including Hyperledger Besu, for its clearing and settlement operations, particularly for commodity tokenization platform

Webinar : [JSCC Besu Production Launch and its Next Steps](#)

DTCC launched a next-generation digital platform to improve global collateral mobility and efficiency, marking the first institutional use of AppChain infrastructure for decentralized finance



Key Highlights

- Built on Besu, offering enterprise-grade privacy, security, and control
- Leverages DTCC AppChain and ComposerX for open, modular deployment of digital applications
- Demonstrates real-time tokenized collateral transfers across traditional and digital assets

Business Value

- Capital Efficiency: Increases liquidity and reduces collateral drag
 - Operational Agility: Automates complex collateral workflows via smart contracts
- Market Connectivity: Bridges traditional and digital asset infrastructures

Why Besu?

- Open-source, Ethereum-compatible enterprise-grade blockchain hosted at neutral foundation under the Linux Foundation
- Only enterprise EVM with complete parity with Ethereum mainnet
- Designed for permissioned networks with privacy controls and interoperability
- Supported by the LF Decentralized Trust community for long-term resilience and innovation

“Collateral mobility is the killer app for institutional blockchain.”

— **Dan Doney**, CTO
DTCC Digital Assets

[Read Announcement](#)

HSBC e-HKD+ - Central Bank Digital Currency and Tokenized Deposits Platform

COMPANY



TECHNOLOGY USED



SUMMARY

HSBC's e-HKD+ platform, built on Besu, explores central bank digital currency applications across public blockchains (Arbitrum, Ethereum, Linea, Polygon) and private networks. Developed for Hong Kong Monetary Authority's Project e-HKD+, the platform focuses on privacy-preserving payments using Privacy Enhancing Technologies and decentralized identity solutions for secure digital currency ecosystems.

RESULTS

- Implemented Privacy Enhancing Technologies for secure transactions
- 700+ respondents providing CBDC adoption insights
- Regulatory compliance: Full HKMA Project e-HKD+ participation and completion

HSBC e-HKD+ - Central Bank Digital Currency and Tokenized Deposits Platform

COMPANY



TECHNOLOGY USED



SUMMARY

Bank-issued stablecoin, for Brazilian customers, giving exposure to USD value, via digital tokens, managed under Safr's banking infrastructure, with legal/regulatory advantages (tax, custody, etc.).

RESULTS

- EVM-Compatible Infrastructure –
- Permissioned & Bank-Grade Controls – Operates as a private/permissioned network with governance, auditability, and compliance features tailored for regulated financial institutions.
- Privacy & Programmability – Adds modules like zero-knowledge proofs and programmable payments to ensure transaction confidentiality



Bank-issued stablecoin, for Brazilian customers, giving exposure to USD value, via digital tokens, managed under Safra's banking infrastructure, with legal/regulatory advantages (tax, custody, etc.).

EVM-Compatible Infrastructure –

Permissioned & Bank-Grade Controls –

Operates as a private/permissioned network with governance, auditability, and compliance features tailored for regulated financial institutions.

Privacy & Programmability – Adds modules like zero-knowledge proofs and programmable payments to ensure transaction confidentiality



Safra

Safra Dólar

O primeiro criptoativo com a solidez do Safra e a força do dólar.

[Abra sua conta](#)

[Sou cliente e quero investir >](#)





Digital Currencies

CBDCs and Beyond



Digital currencies

Allow peer-to-peer transactions and cross-border payments

Reduce transaction times and costs by eliminating intermediaries, allowing for faster and more efficient payments.

Make banking services more accessible to unbanked or underbanked populations, providing them with access to financial services and promoting financial inclusion.

[Download ebook >](#)



Bank of Thailand's Project Inthanon with [Besu](#) was one of the first to demonstrate how blockchain can enhance efficiency and support innovations in payments and supply chain financing by leveraging CBDCs.



ECB research highlighted that platforms like [Besu](#) and Hyperledger Fabric, among others, were "fully interoperable with existing fiat systems." [Read the full report here](#) and [the findings here](#).

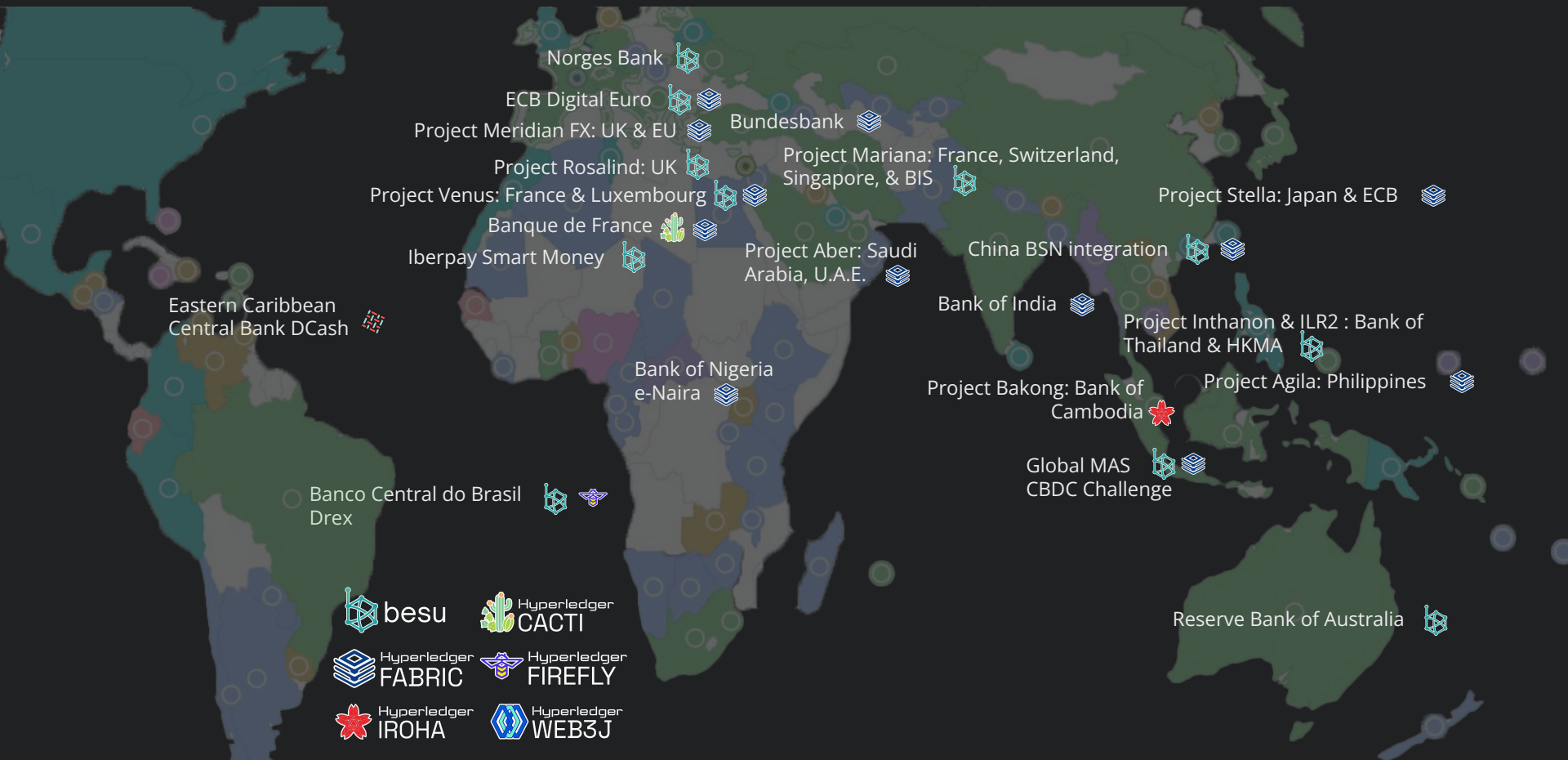


Brasil selected [Besu](#) for its CBDC projects.



Swift CBDC sandbox. 38 central and commercial banks. Hyperledger Firefly + [Besu](#) + Hyperledger Fabric + Quorum and Corda

CBDC projects and experimentations with LF Decentralized Trust tech around the world*





Public Permissioned Public Infrastructure Projects



Public-permissioned networks



LACChain: "Largest permissioned public blockchain". General purpose infrastructure that enables the development of the blockchain ecosystem in Latin America and the Caribbean.



LACCHAIN



The first public sector blockchain services in Europe



Alastria is a neutral public permissioned multi-sector blockchain for regulation-aligned innovation and development



Improves citizen services and provide better transparency on government spending.

Establishing a New Foundation for Trust: How Besu Helps Governments Meet Citizen Needs While Rebuilding Credibility



[Read Case Study](#)





LACChain: A New Blockchain Ecosystem

01 — LACChain is a permissioned public blockchain sponsored by IDB Lab and its partners.

02 — LACChain's approach was to establish a global alliance, build an ecosystem, implement governance, develop economic sustainability tools, and introduce a non-profit orchestration entity.



03 — It was launched in 2019 with the goal to break silos in blockchain usage in Latin America and the Caribbean

04 — The project is the largest permissioned public blockchain infrastructure in the world.

> [Read Case Study LACChain](#)



L2s

Powering Layer 2s



Linea: Built on Besu, optimized
for Linea's Layer 2 rollup.

The Linea logo is displayed in white within a dark, rounded rectangular box. The box has a background of teal and white pixelated patterns. The word "Linea" is in a large, white, sans-serif font, with a small white dot above the final 'a'.

> Read more about linea.build

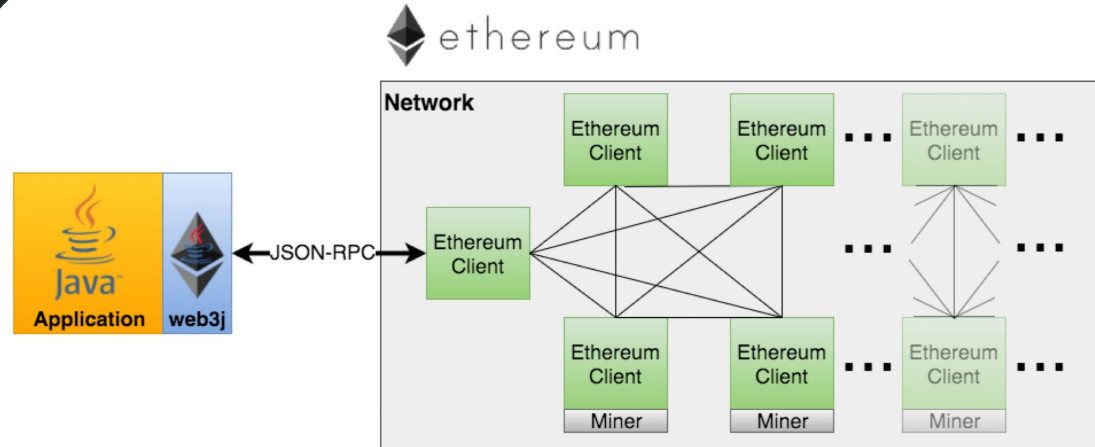
Ethereum related Projects at LFDT





The Java integration library for Ethereum

- Implements the JSON-RPC protocol for communicating with Ethereum nodes
- Performs type conversion for deploying, transacting with and reading data from smart contracts
- Includes wallet and key abstractions for transacting with Ethereum networks



> Watch [Web3J Summit](#)



Hyperledger Cacti is a blockchain integration tool designed to allow users to securely integrate different blockchains.

Cacti is a pluggable, enterprise-grade framework that does not force users to introduce yet another competing blockchain. It aims to provide developers with an abstraction over protocol specific implementations, enabling interoperability.

This enables solutions to more easily adapt to new protocols and make transactions involving multiple public and/or permissioned ledgers.

Focused on cross-chain transaction and atomic swaps.

An “SDK of SDKs” not a “blockchain of blockchains”



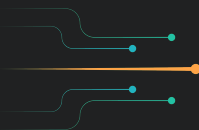
LABS

Yui (IBC)
Harmonia (R3 Corda)



Supernode for web3 apps

Hyperledger FireFly is the first open source Supernode: a complete stack for enterprises to build and scale secure Web3 applications. The FireFly API for digital assets, data flows, and blockchain transactions makes it radically faster to build production-ready apps on popular chains and protocols.





Hiero is an open source, vendor-neutral distributed ledger technology used to build the Hedera public ledger.

Hiero integrates [Besu](#), an open-source Ethereum client, to enhance its smart contract capabilities. By adopting Besu's Ethereum Virtual Machine (EVM), Hiero ensures compatibility with Ethereum-based applications, allowing developers to deploy and manage smart contracts efficiently

Read More: [Introducing Hiero: Bringing Hedera's Core Network Software to Linux Foundation Decentralized Trust](#)

Paladin: Unified Programmable Privacy for EVM

A full stack programmable privacy framework for the EVM that unifies privacy tokens and private EVM workflows in one open runtime. Designed for institutional grade privacy across cash and asset tokens while preserving shared ledger atomicity.

What it enables:

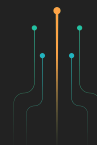
- Atomic composability across transparent tokens, ZKP tokens and private smart contracts.
- Private PvP and DvP workflows with deterministic settlement on a shared EVM ledger.
- Modular onboarding of any privacy token design or privacy group model for scalable enterprise adoption.

 **DECENTRALIZED TRUST**

Paladin, a unified programmable privacy framework for EVM, becomes newest LF Decentralized Trust project

 **PALADIN**

[Read Full
Announcement](#) ➤



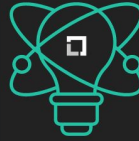


LABS

A flexible space for innovation and testing

A Lab offers an environment where you can easily start work without needing to establish an official project. This option is ideal for:

- Early-Stage Code: Perfect for newly developed code that is still evolving
- Experimentation: Test new frameworks or modules to explore innovative solutions
- Demo Code: Develop and refine demonstration code for showcasing capabilities
- Hackathon Outputs: Transform hackathon ideas into more polished versions
- Research Projects: Advance academic or experimental research into practical applications
- Internship Projects: Provide a platform for interns to contribute meaningful code that can evolve
- Path to Project Status: For code that aims to progress into a formal project in the future



100+



Total number of labs

50+



Active labs

300+



Lab maintainers

Projects with related Labs:

Anoncreds: 1

Aries: 6

Besu: 2

Cacti: 2

Fabric: 42

Indy: 1

Labs That Have Become Projects:

Cacti, FireFly (Graduated)

Bevel, Identus, Solang (Incubating)

Growth in Labs:

18 new labs in 2022

17 new labs in 2023

14 new labs in 2024

22 new labs in 2025

Nightstream

NightStream is an open-source initiative developing a post-quantum zero-knowledge proof system that balances performance, security, and practical deployment constraints. The lab focuses on creating a transparent, lattice-based architecture that provides realistic performance improvements while maintaining post-quantum security guarantees.

[Github >](#)

zkbk

zkbk is a library for building first-person credentials, based on zero-knowledge proof systems.

The library allows developers to create: 1) signing service for issuers to issue personhood credentials, 2) users to make attestations about other users; 3) users to produce showing proofs to verifiers.

[Github >](#)

Generic zk-SNARKs

Effort led by the Privacy & Scaling Explorations (Ethereum Foundation) zkID team to standardise generic zk-SNARKs.

The specification will also include information about the secure composition of these components, in particular with regard to their soundness and zero-knowledge guarantees.

[Github >](#)

Explore Besu Workshops, Meetups, and Webinars

Discover a comprehensive library of Besu-focused events on LF Decentralized Trust YouTube channel. Access links to recordings of past workshops, meetups, and webinars to dive deep into Besu and connect with the community.

[Visit Besu YouTube channel >](#)

Stay informed about upcoming sessions and enhance your understanding of Besu through this curated collection of content.

[Upcoming workshops and Meetups >](#)

LF DECENTRALIZED TRUST workshop

Besu Plug-ins:
Fork-free Client Modifications
to Extend Besu Use Cases

Justin Florentine
ConsenSys

David Leitner
ConsenSys

Fabio Di Fabio
ConsenSys

Date: Tues Oct 29 Time: 8am PT, 11am ET, 4pm CET

consensys



Navil
Rodrigues

How To Prepare For
The Hyperledger
Fabric and Besu
Certification Exams

LF DECENTRALIZED TRUST

member LF DECENTRALIZED TRUST webinar

Simplifying Network
Infrastructure
Management of
Besu

Sankalp Sharma
Co-founder & VP Engineering
Zeeve

Date: October 30 Time: 7AM PT/10AM ET/3PM CET/7:30 PM IST

zeeve

LF DECENTRALIZED TRUST workshop

Configure, Deploy,
and Manage a Besu
Private Network

Jorge Montejó
ioBuilders

Juan Martínez
ioBuilders

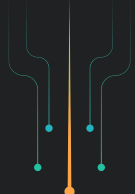
Guillermo Varela
ioBuilders

Marcos Serradilla
ioBuilders

Alejandro Dovale
ioBuilders

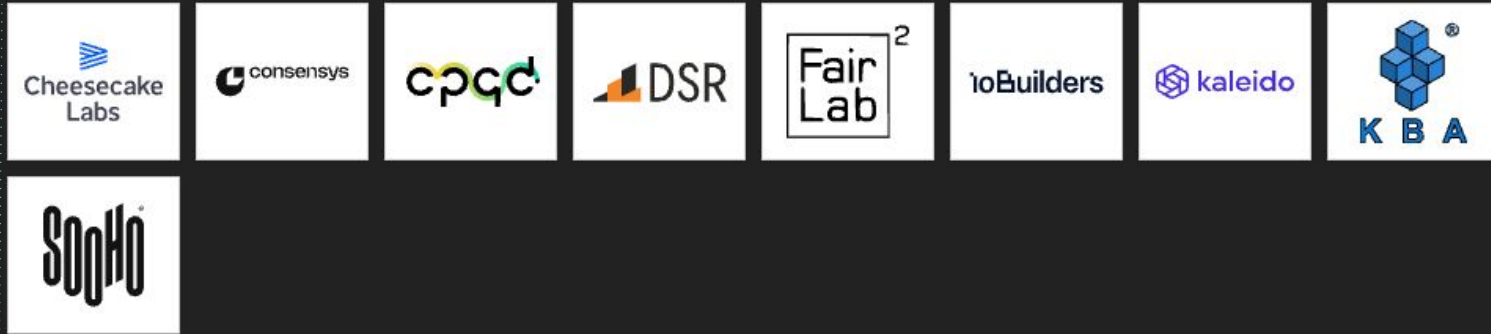
Date: Weds March 26th Time: 8am PT, 11am ET, 4pm CET

ioBuilders



DLF DECENTRALIZED TRUST

CERTIFIED SERVICE PROVIDER



Gain recognition and unlock opportunities
as a trusted Besu service provider

[Become a CSP >](#)

Project updates

Q2 2025

[review project wiki pages for additional updates](#)

Public

Prague Fork

- Prague: polishing ✓
 - Besu on Devnet5 ✓
 - Besu on Devnet6 ✓
 - 25.2.0 is testnet-ready for Prague

Ongoing performance work

- Improve debug_trace* calls performance and reduce output size #8103 ✓
 - debug_traceBlock 3x performance improvement
 - Don't trace memory/stack/storage by default
 - Exclude zeros and empty values
- Block processing
 - Improve conflict detection in parallelization by considering slots to reduce false positives #7923 ✓

Syncing time/memory improvements

- Disable tx indexing during sync (snap/checkpoint) - improves sync time ~ 20% #8181 ✓
- Sync block body memory improvements - reduces memory during syncing #8242 ✓

Client Evolution

- Fusaka planning/dev
 - EOF devnet0
- Bonsai-friendly Archive Mode
- Plug-in technical docs
- Codebase cleanup for better multi-use-case

Enterprise

Move QBFT snap sync out of experimental

- Has been out since 24.7.1 ✓

Bonsai archive

- See PR 7475
- Provides equivalent function to FULL sync FOREST DB

Bonsai archive with state proofs

- Build on PR 7475 to provide proofs for historic state

Stability under load

- Performance and behaviour when the sequenced transaction pool hits its limits
- Most testing until now has been performance close to the TX pool limits - not beyond them

Run enterprise acceptance test every release

Public

4444s and Statelessness

- EIP-4444 history expiry
- Light client exploration
- Verkle Trie ongoing development

Client Evolution

- Fusaka dev/test/ship
 - EOF devnet0
- Removal of enterprise features per sunset plan
- Plug-in technical docs
- Besu on more Layer 2 networks
- Besu as a customizable L2/L3 sequencer
- Ongoing performance work
 - Tx parallel processing: further optimization

Enterprise

Move empty block periods out of experimental

- Has been in out since 24.7.1 

Ensure Verkle + Archive has a solution

- Enterprise options for history expiry
- L1 will start dropping block bodies and receipts from 2Q2025
- Enterprises will need to understand the options Besu provides and the architectures that are available to them
 - Might involve e.g. option to automatically remove blocks and receipts for chainHead-N blocks for non archive, non-full nodes

Removing deprecated features including private transaction and tessera support

Public

Client Evolution

- Modularity of the protocol schedule
- Modularity everywhere

Ongoing Protocol Research

- Besu as an Ethereum reference client in Java
- Statelessness State Expiry research

Enterprise

Modularity work

- Move consensus logic out of core Besu monorepo
- Allow faster iteration on both core Besu and enterprise Besu features

BFT peering via non-validators

- Allow edge-of-network nodes that aren't BFT validators

[Check the Zenhub ➤](#)

Use case tracker:

Explore 300+ use cases built with LF Decentralised Trust technologies (50+ with Besu)

Explore tracker ➤

 NORGES BANK Bank of Norway's CBDC experience Organization: Bank of Norway Description: In 2018, a Norges Bank working group presented an overview of topics that should be emphasized in an assessment of whether Norges Bank... Project: Besu Sector/Industry: Financial Services CBDC Public Sector Status: In-Development Website: https://www.norges-bank.no/en/topi... News and Reports: https://www.euronews.com/next/202... Case Studies: ... LF Decentralized Trust Membership: LFD Associate Member	 BANCODE ESPAÑA Eurosisistema Spain's smart money Organization: Bank of Spain BBVA Caixa Bank B Description: The Spanish financial sector completed the Smart Money experiment on the technical aspects of a digital euro's distribution, use, ... Project: Besu Sector/Industry: Financial Services Public Sector Go Status: In-Development Website: https://www.iberpay.es/media/21555... News and Reports: https://www.lidgerinsights.com/spa... Case Studies: ... LF Decentralized Trust Membership: Not a member	 HYPERLEDGER Capital Markets SPECIAL INTEREST GROUP eThaler Project Organization: Hyperledger Foundations Labs Description: eThaler was created by Capital Markets Special Interest Group, with the aim to explore the process of modeling a CBDC using a token ... Project: Besu Sector/Industry: CBDC Financial Services Status: In-Development Website: https://wiki.hyperledger.org/display... News and Reports: https://www.forbes.com/sites/micha... Case Studies: ... LF Decentralized Trust Membership: N/A	 BAXE BAXE - The New Standard of ... Organization: Baxe Description: BAXE is an integrated communications and peer-to-peer exchange platform, leveraging blockchain technology to create a ... Project: Besu Sector/Industry: Financial Services Digital Identity Status: In-Production Website: https://baxe.com/ News and Reports: https://medium.com/@baxe_app/the... Case Studies: ... LF Decentralized Trust Membership: Not a member	 BIS Project Genesis: Green Bond ... Organization: BIS Innovation hub Singapore Digital Description: The BIS Innovation Hub and the Hong Kong Monetary Authority (HKMA) successfully developed two prototype digital platforms that bring to life the ... Project: Besu Hyperledger Fabric Sector/Industry: Climate Action Financial Services T Status: In-Production Website: https://www.bis.org/about/bis/h/topic... News and Reports: ... Case Studies: ... LF Decentralized Trust Membership: Not a member
 Inthanon-LionRock Inthanon-LionRock (mCBDC ... Organization: BIS Innovation hub Singapore Conso Description: A prototype of multiple Central Bank Digital Currencies (mCBDCs) led by the Bank for International Settlements Innovation Hub, Hong Kong Monetary ... Project: Besu Sector/Industry: CBDC Public Sector Financial Servi	 RESERVE BANK OF AUSTRALIA Project Atom - CBDC research... Organization: ConsenSys The Reserve Bank of Aus Description: Bank of Australia partnered with ConsenSys to develop a POC for the issuance of a tokenized form of CBDC that can be used by wholesale mark... Project: Besu Sector/Industry: Financial Services CBDC Governme	 DTCC DTCC's Post-Trade infrastruc... Organization: Depository Trust & Clearing Corporati Description: DTCC is exploring a number of different applications of blockchain within financial space using different blockchain networks, including ... Project: Besu Hyperledger Fabric Sector/Industry: Financial Services Digital Assets Ac	 European Central Bank ECB - CBDC experiments Organization: European Central Bank Description: ECB President Christine Lagarde has suggested that a digital euro can complement cash. Furthering the exploration of a digital euro, the ... Project: Hyperledger Fabric Besu Sector/Industry: Financial Services Government Pub	 FNALITY Fnality: Decentralized Financi... Organization: Fnality Description: Fnality is distributed Financial Market Infrastructure built on Hyperledger Besu that interoperates and connects to many existing systems and rails ... Project: Besu Sector/Industry: Financial Services Tokenization Dig

Community calendar: Public meetings

The "Calendar of Public Meetings" lists upcoming community calls and events, helping members stay informed and engaged. It ensures collaboration and transparency by making key meetings accessible, enabling participants to contribute, stay updated, and foster an inclusive environment for active community growth.

[View calendar >](#)

Please note: Migration is in process to [LFX community meetings](#)

LF DECENTRALIZED TRUST

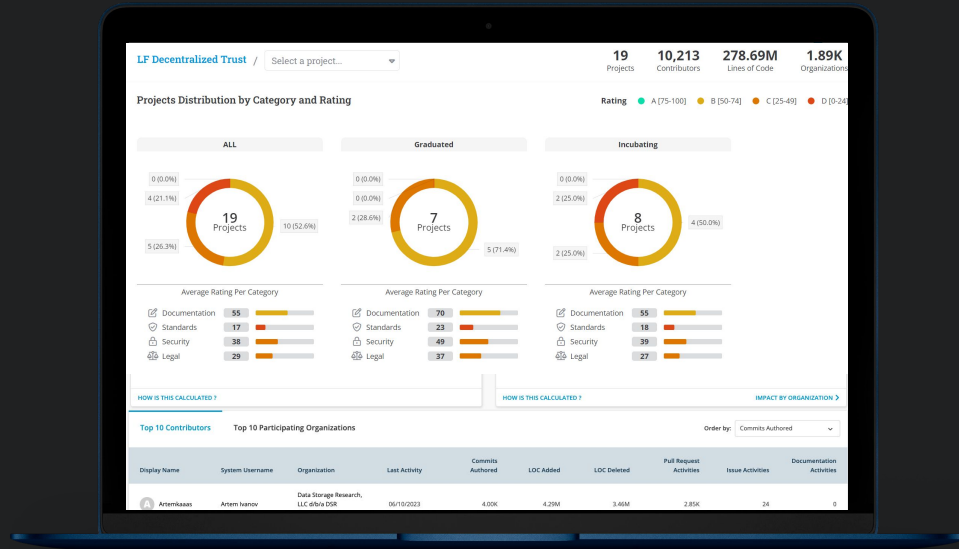
< September 2024 >						
Mo	Tu	We	Th	Fr	Sa	Su
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

Projects 20 projects X

Search Projects

- ☒ **LF Decentralized Trust** ●
- ☒ AnonCreds ●
- ☒ Besu ●
- ☒ Caliper ●
- ☒ Hiero ●

Data-driven insights enables you to make informed decisions



Project group overview, aggregate statistics and project cards.

➔ Project Analytics

"LFX Insights has enabled me as both the CTO in my organization, as well as the Governing Board chairperson, to better see the growth of the hosted projects as well as my organization's impact"



Len Santalucia

CTO, Vicom Infinity and
Governing Board chairperson for
the Open Mainframe Project

Features

- ✓ Multi-Dimensional Reporting
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- ✓ Contributor Affiliations
- ✓ Shared Innovation
- ✓ Identity Management
- ✓ Telemetry from Multiple Data Sources
- ✓ Community Contributor Boards
- ✓ Marketing Metrics

Get trained and certified: Besu Certified Professional

Web3, and decentralized finance (DeFi) continue to expand. With the new certification, you are well-equipped to capitalize on the potential of the Ethereum network, enabling you to innovate and streamline operations through smart contracts and decentralized applications.

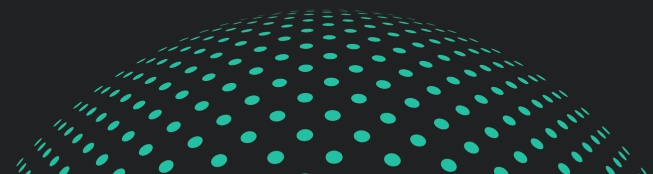
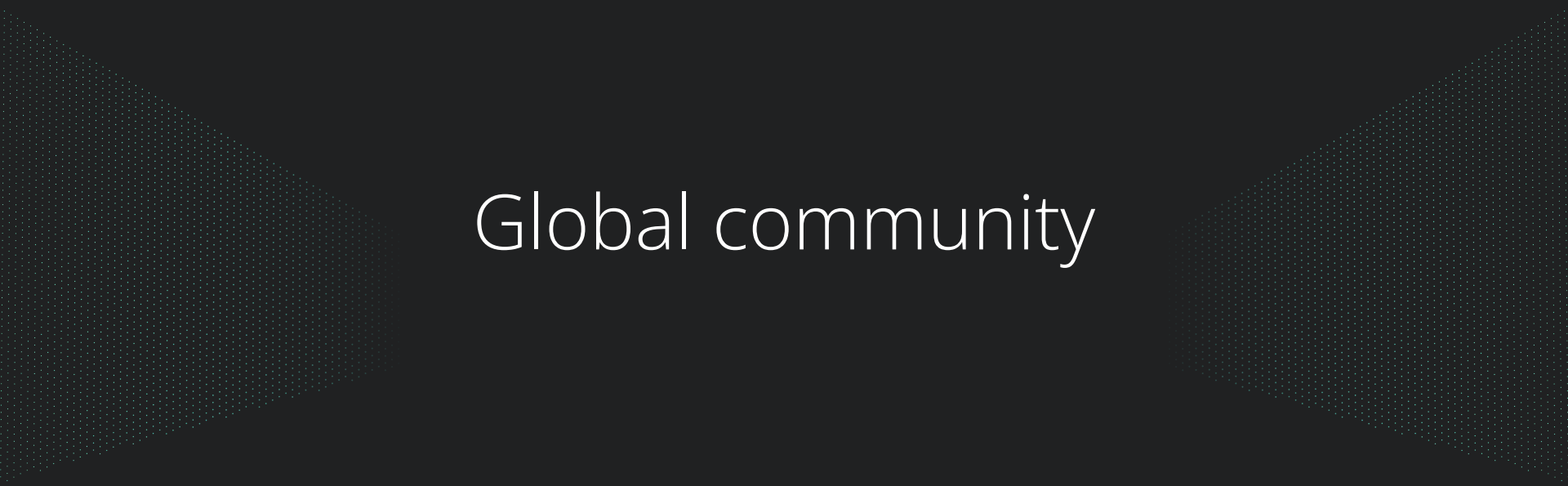
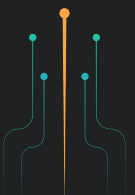
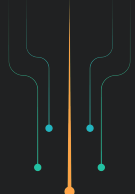
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Now LIVE!



Other LF Decentralized Trust courses:

- Blockchain: Understanding Its Uses and Implications
- Introduction to LF Decentralized Trust Technologies
- Introduction to Sovereign Identity Blockchain Solutions: Indy, Aries & Ursa
- Becoming a Hyperledger Aries Developer
- Hyperledger Fabric Fundamentals
- Besu Essentials: Creating a Private Blockchain Network



Global community

DTCC**OpenAssets****Shielded****19**

PROJECTS

1.2M

CONTRIBUTIONS

394K

TOTAL COMMITS



Associate members





Global events & thought leadership

Driving enterprise adoption

LF Decentralized Trust community thought leaders present at various industry events around the world, demonstrating their ongoing commitment to global collaboration, open dialogue, and innovation in decentralized tech.



Join discord

Explore project channels, special interest groups, working groups, and regional chapters. Engage with contributors, share ideas, and stay connected.

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4,700 users
67 Core Project Maintainers





Ready to shape the future of decentralized technologies?

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Any questions?

Our leadership team is ready to assist



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